Fit an sjSDM Model

Description

Fits a joint Species Distribution Model (jSDM) using the sjSDM package with specified abiotic, spatial, and error family configurations.

Usage

```
fit_jsdm_model(
  data_to_fit = NULL,
  sel_abiotic_formula = NULL,
  abiotic_method = c("linear", "DNN"),
  sel_spatial_formula = as.formula(~0 + coord_long:coord_lat),
  spatial_method = c("linear", "DNN", "none"),
  error_family = c("gaussian", "binomial"),
  device = c("cpu", "gpu"),
  parallel = 0L,
  compute_se = FALSE,
  biotic = NULL,
  alpha_cov = 0.5,
  alpha_coef = 0.5,
  alpha_spatial = 0.5,
  lambda_cov = 0,
  lambda_coef = 0,
  lambda_spatial = 0,
  ...,
  iter = 100L,
  n_early_stopping = NULL,
  verbose = FALSE,
  step_size = NULL
)
```

Arguments

`data_to_fit`	A list containing the data to fit the model. Must include: - 'data_community_to_fit': matrix of community composition (constant-presence taxa already removed by 'filter_constant_taxa()') - 'data_abiotic_to_fit': data frame of abiotic variables, already scaled by 'scale_abiotic_for_fit()' - 'data_spatial_to_fit': data frame of spatial predictors, already scaled by 'scale_spatial_for_fit()'. Required only when 'spatial_method' is not '"none"'.
`sel_abiotic_formula`	A formula object specifying the abiotic (environmental) predictors
`abiotic_method`	Method for modeling abiotic effects. One of "linear" (default) or "DNN" (deep neural network)
`sel_spatial_formula`	A formula object specifying the spatial predictors. Defaults to an interaction between longitude and latitude ('~ 0 + coord_long:coord_lat'). Only used if 'spatial_method' is '"linear"' or '"DNN"'.
`spatial_method`	Method for modeling spatial effects. One of '"linear"' (default), '"DNN"' (deep neural network), or '"none"' (no spatial structure)
`error_family`	Error family distribution. One of "gaussian" (default) or "binomial". If "binomial", community data is converted to presence/absence and a probit link is used
`device`	Computing device to use. One of "cpu" (default) or "gpu"
`parallel`	Number of CPU cores to use for parallel processing. Only applicable if 'device = "cpu"'. Default is '0L' (no parallelisation).
`compute_se`	Logical indicating whether to compute standard errors inline during model fitting. Default is 'FALSE'. Prefer the post-hoc approach via 'compute_jsdm_se()' in a separate pipeline target, which allows CPU parallelisation independent of the GPU device setting.
`biotic`	Optional biotic structure passed to 'sjSDM::sjSDM()'. Defaults to 'NULL', which creates 'sjSDM::bioticStruct()' from the supplied 'alpha_cov' and 'lambda_cov'. Use 'sjSDM::bioticStruct(diag = TRUE)' to fit a no-associations variant.
`alpha_cov`, `alpha_coef`, `alpha_spatial`	Elastic-net mixing parameters for biotic association covariance, abiotic coefficients, and spatial coefficients. Each must be between zero and one, inclusive.

`lambda_cov`, `lambda_coef`, `lambda_spatial`	Non-negative regularization strengths for biotic association covariance, abiotic coefficients, and spatial coefficients.
`...`	Additional arguments passed to 'sjSDM::sjSDM()' (e.g. 'sampling', 'seed'). Do NOT pass 'iter', 'control' or 'early_stopping' directly — use the dedicated 'iter' and 'n_early_stopping' parameters instead.
`iter`	Positive integer. Number of training epochs. Default is '100L'.
`n_early_stopping`	Early stopping patience. Controls how many consecutive epochs without improvement are tolerated before training is halted. Three accepted values: - 'NULL' (default): auto-compute as 'round(iter * 0.20)', i.e. at least 20 % of the epoch budget. - '0L' or negative integer: disables early stopping entirely. - Positive integer: uses 'max(value, round(iter * 0.20))' to ensure patience is never set below 20 % of 'iter'. Passed as 'early_stopping_training' in 'sjSDM::sjSDMControl()'.
`step_size`	Optional positive integer training batch size. ‘NULL' retains sjSDM’s default except for fewer than ten training rows, where batch size one prevents its automatic ten-percent batch from rounding down to zero.

Details

This function prepares the data and fits a joint Species Distribution Model using the sjSDM package. The spatial structure is modeled using an interaction term between longitude and latitude coordinates. When binomial error family is specified, the community data is converted to binary presence/absence data.

Standard error computation ('compute_se = TRUE') may fail with certain model configurations, particularly when using DNN methods or complex spatial structures. If SE computation fails, the model will still be returned with a warning.

Value

An object of class sjSDM containing the fitted model

See Also

sjSDM::sjSDM, sjSDM::linear, sjSDM::DNN, compute_jsdm_se